

Algorithms Problem Solving

Level 3

Update Client by Account
Number



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Update Client by Account Number

Definition

Key-Based Record Update

 Program locates an existing client using its Account Number and modifies selected fields.

 The updated collection is then written back to the file so the changes become persistent.

Explanation

Record Lookup

-  The client is located using its unique Account Number.

-  The Account Number acts as the stable key used to identify the record.

User Confirmation

-  The program can display the existing record and ask the user to confirm the update.

-  This prevents accidental modification of stored data.

Reference-Based Modification

-  The matching `sClient` object is modified directly through a reference.

-  Fields such as `PinCode`, `Name`, `Phone`, and `AccountBalance` can be replaced with new values.

Persistent File Update

-  After modification, the complete client vector is saved back to the file.

-  Rewriting the file ensures that the updated values remain available after the program terminates.

Primary Key Preservation

-  `AccountNumber` remains unchanged because it is the identifier used to locate the client.

-  Updating ordinary attributes should not unintentionally change the record's identity.

Procedures

1. Load all clients from the file.
2. Read the Account Number to update.
3. Search for the matching client.
4. Confirm that the user wants to update it.
5. Modify the selected fields.
6. Preserve the original Account Number.
7. Save the complete vector back to the file.
8. Display a successful update message.

💡 Example

Enter Account Number to Update:

A150

Client Found!

Enter New Details:

Name:

Mohammed Updated

Balance:

7500

➡ Result

Client Updated Successfully!

Note

- ◆ The Account Number functions as the record's identifying key.
- ◆ The vector must be modified before saving the new data to disk.
- ◆ The same serialization logic used when adding clients can be reused when rewriting updated records.

⊘ Warning

- ◆ Use a reference such as ``for (sClient& C : vClients)`` when modifying vector elements directly.
- ◆ Do not accidentally modify a local copy such as ``for (sClient C : vClients)``.
- ◆ Changing the Account Number can break the relationship between the lookup key and the stored record.
- ◆ Updating the in-memory vector without saving it back to the file does not make the change persistent.



Comparison

▣ `UpdateClientByAccountNumber()`

- ✓ Locates an existing record using its key.
- ✓ Changes selected fields while preserving the Account Number.
- ✓ Rewrites the updated collection to persistent storage.

▤ `DeleteClientByAccountNumber()`

- ✓ Locates a record using its key.
- ✓ Removes or marks the record for deletion.
- ✓ Rewrites the file without the deleted record.

✦✦ Summary

- ✓ Searches for a client using Account Number.
- ✓ Updates the selected fields through a reference.
- ✓ Preserves the identifying Account Number.
- ✓ Saves the modified records back to persistent storage.

Thank you 😊